

Biology All In One (BAIO) Learning System

PU-II Biology · Narendra Sir – M.Sc (Biochemistry), B.Ed, KSET

Chapter 9: Biotechnology – Principles and Processes

Worksheet · Cloning Vectors & pBR322

I. Multiple Choice Questions

- Plasmids are described as:
(A) Linear, non-replicating DNA (B) Autonomously replicating, circular, extra-chromosomal DNA
(C) Single-stranded RNA molecules (D) Protein-based vectors
- Which two types of molecules are commonly used as cloning vectors?
(A) Plasmids and bacteriophages (B) Ribosomes and mitochondria
(C) Histones and nucleosomes (D) tRNA and rRNA
- 'ori' stands for:
(A) Organism of replication (B) Origin of replication
(C) Order of insertion (D) Original recombinant identity
- In pBR322, which two genes serve as selectable markers?
(A) ampR and tetR (B) rop and ori
(C) EcoRI and BamHI (D) lacZ and lacY
- Bacteriophages, compared to plasmids, generally have:
(A) A lower copy number (B) A higher copy number
(C) No copy number at all (D) The same copy number always

II. Fill in the Blanks

- Plasmids and _____ are the most commonly used cloning vectors.
- The site from which vector replication begins is called the _____.
- In pBR322, ampR and tetR serve as _____ genes.
- Cloning sites are the recognition sites for _____ present on a vector.
- The rop gene in pBR322 codes for proteins involved in _____ of the plasmid.
- Plasmid copy number can range from 1–2 up to _____ copies per cell.

III. Match the Following

Column A	Column B
1. ori	a. High genome copy number vector
2. amp ^R /tet ^R	b. Restriction enzyme recognition site
3. Cloning site	c. Origin of replication
4. Bacteriophage	d. Selectable marker gene

IV. True / False

- Plasmids are autonomously replicating, circular, extra-chromosomal DNA. [True / False]

2. Bacteriophages generally have a lower genome copy number than plasmids. [True / False]
3. A high copy number ori generally increases the yield of recombinant protein. [True / False]
4. pBR322 has no selectable marker genes. [True / False]
5. Cloning becomes complicated when more than one recognition site is present on the vector. [True / False]

V. 2 Marks Questions

1. What is the role of the origin of replication (ori) in a cloning vector?

2. Why is ligation in pBR322 carried out only at restriction sites within the antibiotic resistance genes?

VI. 3 Marks Questions

1. List the key components of the pBR322 cloning vector.

2. Differentiate between the copy number of plasmids and bacteriophages.

VII. 5 Marks Questions

1. With reference to the plasmid vector pBR322, explain the role of the origin of replication (ori), selectable marker genes, and cloning sites in the construction of recombinant DNA. Draw and label a simple diagram of pBR322.
